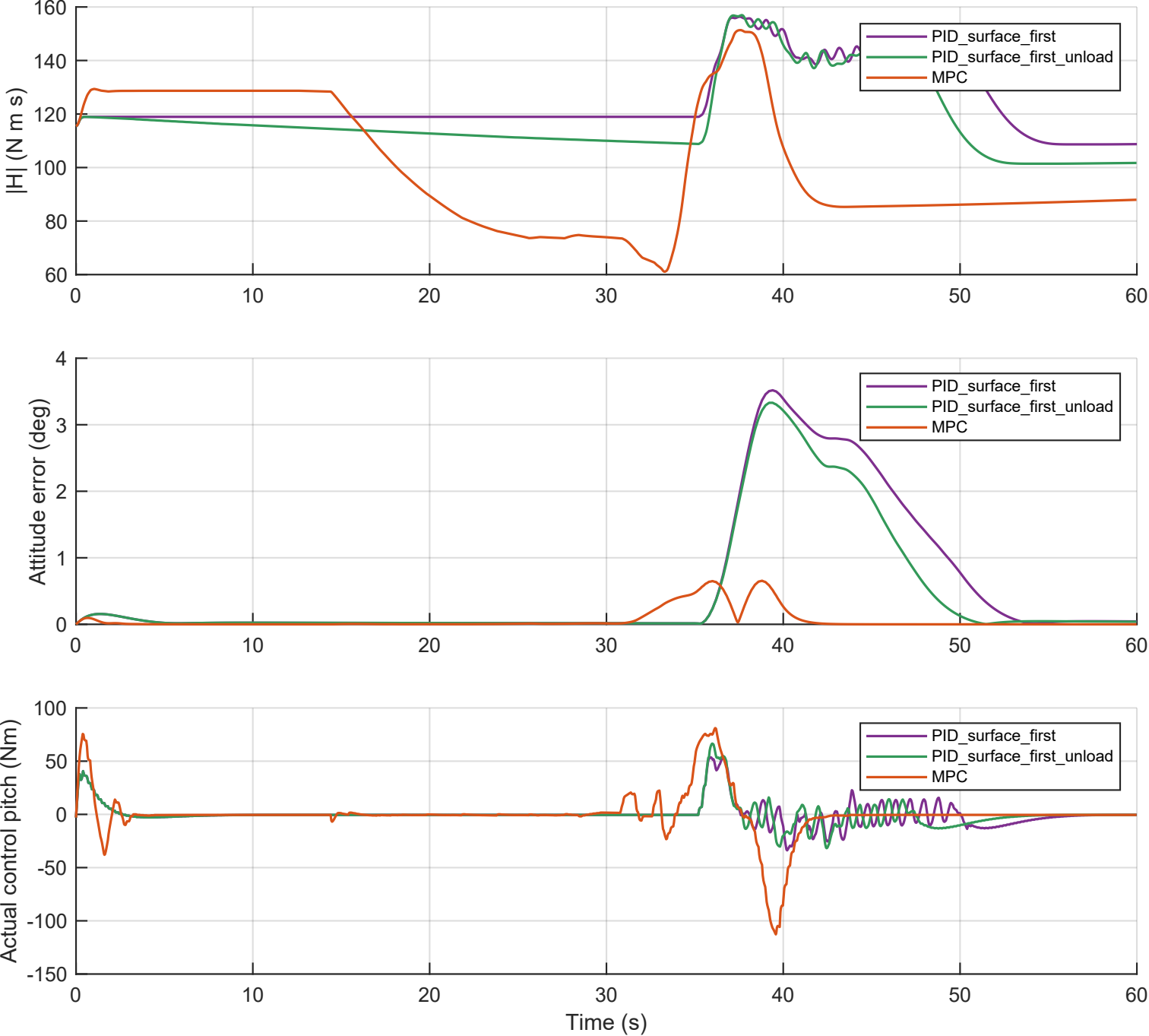

|H| (N m s)
160
140
120
100
80
60
0
10
20
30
40
50
60
PID_surface_first
PID_surface_first_unload
MPC
Attitude error (deg)
4
3
2
1
0
0
10
20
30
40
50
60
PID_surface_first
PID_surface_first_unload
MPC
Actual control pitch (Nm)
100
50
0
-50
-100
-150
0
10
20
30
40
50
60
Time (s)
PID_surface_first
PID_surface_first_unload
MPC